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AI Engineer

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Hiring Team
Smiths Detection
Wiesbaden

6 November 2026

Application for Software Engineer - Computer Vision & Artificial Intelligence

Dear Hiring Team,

I am writing to apply for the Software Engineer - Computer Vision & Artificial Intelligence position at Smiths Detection. I have a Master's in Artificial Intelligence and three years of experience building and deploying AI applications, and I would like to put the machine learning side of that work into imaging products.

My most image-like work so far is time-series modelling. At Covestro I built and compared Random Forest, XGBoost and LSTM models on sensor data to predict synthesis outcomes, engineered the features, and benchmarked the results against empirical curve fitting. The thesis that followed was about evaluation: I set up automated generation and retrieval tests with RAGAS and PytrecEval against ground truth supplied by chemists. At Kniit I raised precision and recall from 33% and 45% to 93% and 90% by measuring every change against a ground truth derived from SAP data. Preparing and checking data and judging a model by numbers is how I work.

On the software side I write Python every day. I have built FastAPI backends, containerised them with Docker, deployed to AWS with CDK and run CI/CD in GitLab. I also work with PyTorch and TensorFlow, and I fine-tuned an open-source LLM with QLoRA.

I should be direct about the gaps. I have not worked professionally on classical 2D or 3D image processing, I have not written C++, and I have never captured images on X-ray equipment. My image work so far is limited to CNN and ResNet image classification on MNIST and Fashion-MNIST in my studies. I learn a new domain quickly from the people who work in it, as I did with chemists and SAP experts.

Smiths Detection builds X-ray and CT screening systems and, through iCMORE, adds automatic detection of prohibited items on top of them. Work where an algorithm has to be validated on real scanner images before it ships is the kind I want to do.

I am based in Wiesbaden, hold an EU Blue Card and can start on 15 November 2026. My English is business fluent and my German is at B1 level. Thank you for considering my application.

Kind regards,

Chandu Prasad Bhairapu